



Blessy Edison

M.Tech Embedded Systems

9488128754 | blessyedison14@gmail.com

<https://www.linkedin.com/in/blessyedison11/> | <https://github.com/blessyed>

Professional Summary

M.Tech (Embedded Systems) student with practical experience in **STM32 firmware development**, **RTOS concepts**, sensor interfacing, and industrial embedded systems. Completed a 6-month **R&D internship at Kawasaki Robotics** focused on embedded firmware, **PCB design**, and real-time sensor monitoring. Seeking internship opportunities in embedded systems for automotive, industrial automation, and real-time applications.

Core Skills

Programming: Embedded C, Advanced C, C++, Python

RTOS : Task scheduling, Inter-task communication, Synchronization

Microcontrollers: STM32 (M4/M7), PIC16F877A, ESP32

Hardware & PCB: Sensor interfacing, schematic design, PCB layout (KiCad)

Communication & Interfaces: CAN, UART, SPI, I2C, TCP/IP socket programming

Visualization: PyQt GUI development

Tools: STM32CubeIDE, Keil uVision, KiCad, MATLAB, Unity, GitHub

Industrial Experience

Kawasaki Robotics — R&D Intern

Semiconductor Wafer Handling Division

Jan 2024 – Jun 2024

- Worked in the R&D team on semiconductor wafer-handling robotic systems used in industrial automation.
- Implemented long-distance wafer detection using **Time-of-Flight (CH101) sensors** interfaced with **STM32 microcontrollers**.
- Developed **bare-metal STM32 firmware** for sensor acquisition, calibration, and real-time processing
- Designed a custom **KiCad PCB** integrating controller, sensor, and power circuitry, while implementing **TCP/IP socket** communication and a **Python PyQt GUI** for real-time data transfer, visualization, and monitoring.

Projects

RTOS-Based Collision Avoidance System

Jan 2026 – April 2026

STM32 - Embedded C - RTOS - CAN

- Designed a real-time collision avoidance system using RTOS task scheduling.
- Implemented concurrent tasks for sensor acquisition, decision logic, and actuator control.
- Used inter-task communication and synchronization to ensure deterministic response.
- Targeted for automotive and industrial safety-critical applications.

IoT-Based Digital Twin Conveyor Belt Monitoring System

Jan 2026 – April 2026

Unity - ESP32 - Sensors - IoT - Embedded C

- Developed conveyor belt monitoring using ESP32 and sensors for real-time condition sensing.
- Created a digital twin in Unity for virtual conveyor visualization.
- Enabled real-time monitoring of overload and jam conditions through hardware-to-digital twin data transfer.

Vehicle-to-Vehicle Brake Failure Alert System

Jan 2026 – April 2026

ESP32 – LoRa - GPS - Embedded C

- Developed a vehicle-to-vehicle communication prototype for transmitting brake failure alerts between vehicles.
- Implemented long-range wireless communication using LoRa with ESP32 for real-time safety message transfer.
- Integrated GPS-based location sharing to support early warning and collision prevention in automotive safety applications.

Wearable Health Monitoring System

Aug 2026 – Dec 2026

Embedded C – STM32

- Developed a wearable health monitoring system using MAX30102, temperature sensor, and accelerometer with STM32.
- Implemented real-time monitoring of heart rate, SpO₂, temperature, and motion parameters using Embedded C

Education

Amrita Vishwa Vidyapeetham, Bengaluru

M.Tech – Embedded Systems

2025 – Present

Coimbatore Institute of Engineering and Technology, Coimbatore

B.E – Electrical & Electronics Engineering

2020 – 2024

Training & Certifications

Advanced Embedded Systems Professional Training (ECEP) – Emertxe Information Technologies, Bengaluru

- Built strong proficiency in **C, C++, Data Structures, and Algorithms (DSA)** through extensive programming practice.
- Developed solid theoretical understanding of **Embedded Linux** concepts.
- Implemented multiple embedded projects using **PIC microcontrollers**

Achievements

- **President, Department of Electrical and Electronics Engineering (2023–2024)**
- **Finalist – Niral Thiruvizha Project Competition (2024)**
- Published a technical research paper on fault detection in power electronics systems
- Completed a **6-month industrial R&D internship at Kawasaki Robotics**